

FSK116: Tutorial Problem Numbers / Tutoriaal Probleem Nommers

2017

1 Week: 13 February 2017 / 13 Februarie 2017

All the indicated numbers refer to the “Problems” section in Serway; e.g. P1.9 is problem 9 from chapter 1.

Alle nommers verwys na die “Problems” afdeling in Serway; bv. P1.9 is probleem 9 uit hoofstuk 1.

P1.9 (a) Not dimensionally correct (b) Dimensionally correct

P1.12 $\frac{\text{m}^3}{\text{kg} \cdot \text{s}^2}$

P1.19 Number of pages are insufficient

P1.25 $r_{\text{Al}} = 2.86 \text{ cm}$

P1.26 $r_{\text{Fe}}(1.43)$

P1.30 (a) 2.07 mm (b) 8.62×10^{13} times as large

P1.62 10^{11} stars

P1.66 (a) $478 \text{ cm}^3/\text{s}$ (b) 0.225 cm/s (c) explanation

P2.5 (a) 2.30 m/s (b) 16.1 m/s (c) 11.5 m/s

P2.7 (a) -2.4 m/s (b) $\approx -3.8 \text{ m/s}$ (c) 4.0 s

P2.21 (a) 2.00 m (b) -3.00 m/s (c) -2.00 m/s^2

P2.25 (a) $4.98 \times 10^{-9} \text{ s}$ (b) $1.20 \times 10^{15} \text{ m/s}^2$

P2.30 (a) 20.0 s (b) 1000 m is needed to land (c) The plane can not land.

P2.75 (a) $v_B = \left(\frac{1}{\tan \theta}\right) v$ (b) Explanation